Eesha Tolety

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CAREER OBJECTIVE

I'm a graduate student in aerospace engineering, passionate about structures, fluids, and materials. I lead the structural design and have been part of the fabrication and testing team for an exciting VTOL drone project at CU Boulder. I'm eager to apply my knowledge in simulation, CAD, and prototyping to contribute to an engineering team that tackles impactful and innovative challenges in the aerospace industry.

TECHNICAL SKILLS

- **Programming Languages:** MATLAB, MATLAB Simulink, Python, C++,
- **Software:** SolidWorks, ANSYS(Fluent, CFX, Workbench, APDL, Structural), ABAQUS, OpenFOAM, Paraview, Solid-Edge, AutoCAD, Onshape, NX, Rhino, Prusa Slicer, 3-D Printing Slicers.

EDUCATION

- **Master of Science in Aerospace Engineering** **August 2024 - May 2026**
  University of Colorado Boulder *Boulder, CO*
  Fluids, Structures and Materials
  GPA: 3.67/4.00
- *Bachelor of Science in Aerospace Engineering* *July 2019 - June 2023*
  Amity University **Noida, India**
  GPA: 8.74/10.00 Grade: A

PROJECTS

- **FLAIR** **August 2025 - May 2026**
  Graduate Project, University of Colorado Boulder *Boulder, CO*
  This project involves incorporating a new type of payload into a large VTOL drone. As one of the structural leads on the project, I have been using SolidWorks and Solid Edge for the design work. Additionally, I utilized SolidWorks and Abaqus for design analysis. During the production phase, I assisted with fabrication and testing of the components before the final flight.
- **Microstructural and mechanical analysis of multi-walled Carbon Nanotube(CNTs) and Magnesium(Mg) composite.** **January 2023 - June 2023**
  Major Project, Amity University *Noida, India*
  A CNT-reinforced magnesium composite was developed using a modified friction stir processing technique to improve CNT dispersion. XRF confirmed the presence of carbon, magnesium, and oxygen, while SEM showed a consistent distribution of CNTs across the sample. Mechanical properties, such as macro-hardness, were also investigated.
- **Overall Enhancement of Physio-mechanical characteristics of Composite materials produced with Carbon Nanotube (CNTs)** **August 2022 - January 2023**
  Minor Project, Amity University *Noida, India*
  In this study, pure magnesium was reinforced with varying weight percentages (0.3%–1.8%) of multiwalled CNTs, prepared from organic waste like sugarcane, using stir casting. Mechanical properties were significantly improved, with the optimal CNT content found between 1.2-1.5% by weight for overall property enhancement.

EXPERIENCE

- **University of Colorado Boulder** **January 2026 - May 2026**
  Teaching Facilitator: Foundations of Propulsion *Boulder, CO*
  Assisting students in resolving questions related to numerical problems and conceptual understanding. Evaluating homework assignments and examinations. Organizing study halls to facilitate a deeper comprehension of the subject matter.
- **The Digital Media Center** **August 2025 - May 2026**
  Student Leader
  Boulder, CO
  Overseeing student workers, managing communication between staff and management, and improving 3-D printing workflow and services.
  Assistant technician *Boulder, CO*
  Supporting technical operations, equipment troubleshooting, and hands-on fabrication assistance for the 3-D printing lab.
- **Aeromodelling Club** **August 2022 - June 2023**
  Vice-President, Amity University *Noida , India*
  Preparing for competitions and managing inventory for model plane construction. Conducting workshops to share knowledge on designing and flying drones and model planes.

- **Hindustan Aeronautics Limited** **May 2022 - June 2022**
  Internship, Study of In-flight Refueling System *Nasik, India*
  In this study, I gained detailed information about the in-flight refueling system, including the necessary components for the process and how it is executed. Through my internship in a comprehensive shop, I was able to observe the entire operation in detail and gain real-life experience, as well as identify the problems that can arise during this process.

ACHIEVEMENTS

- Tolety, E., Gupta, A., Singh, N., Saluja, R.K., Gahlot, N.K. (2024). Microstructural Analysis of Mg/CNT Surface Composite Using Friction Stir Processing. Springer, Singapore.
  https://doi.org/10.1007/978 − 981 − 97 − 1306 − 6 − 15

INVOLVEMENT

- **Jax Foundation** **August 2022 - December 2022**
  Volunteer teacher, providing basic English, Science, and Mathematics instruction to underprivileged children.
- **Aeromodelling Club** **August 2019 - June 2023**
  I began as a club member and became Vice-President in my final undergraduate year.